1.5.3 Electrostatic potential at a point due to an electric dipole

Consider two equal and opposite charges separated by a small distance $2a$ as shown in Figure 1.25. The point P is located at a distance r from the midpoint of the dipole. Let θ be the angle between the line OP and dipole axis AB.

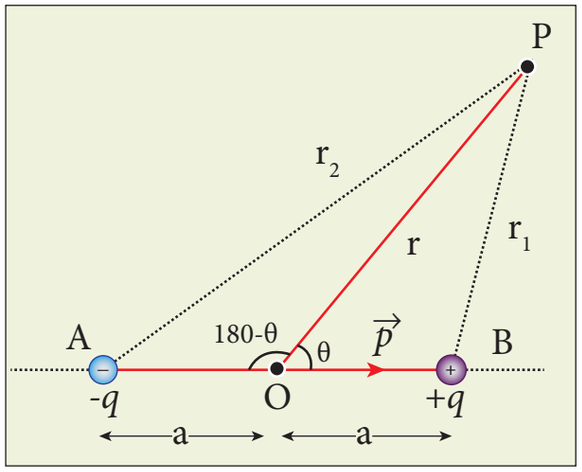


Figure 1.25 Potential due to electric dipole

Let r_1 be the distance of point P from $+q$ and r_2 be the distance of point P from $-q$.

Potential at P due to charge $+q = \frac{1}{4\pi\epsilon_0}\frac{q}{r_1}$

Potential at P due to charge $-q = -\frac{1}{4\pi\epsilon_0}\frac{q}{r_2}$

Total potential at the point P

$$V = \frac{1}{4\pi\epsilon_0} q\left(\frac{1}{r_1} - \frac{1}{r_2}\right) \qquad (1.35)$$

Suppose if the point P is far away from the dipole, such that $r>>a$, then equation (1.35) can be expressed in terms of r.

By the cosine law for triangle BOP

$$r_1^2 = r^2 + a^2 - 2ra\cos\theta$$

$$r_1^2 = r^2\left(1 + \frac{a^2}{r^2} - \frac{2a}{r}\cos\theta\right)$$

Since the point P is very far from the dipole ($r>>a$). As a result the term $\frac{a^2}{r^2}$ is very small and can be neglected. Therefore

$$r_1^2 = r^2\left(1 - 2a\frac{\cos\theta}{r}\right)$$

$$\text{(or) } r_1 = r\left(1 - \frac{2a}{r}\cos\theta\right)^{\frac{1}{2}}$$

$$\frac{1}{r_1} = \frac{1}{r}\left(1 - \frac{2a}{r}\cos\theta\right)^{-\frac{1}{2}}$$

Since $\frac{a}{r} << 1$, we can use binomial theorem and retain the terms up to first order

$$\frac{1}{r_1} = \frac{1}{r}\left(1 + \frac{a}{r}\cos\theta\right) \qquad (1.36)$$

Similarly applying the cosine law for triangle AOP,

$$r_2^2 = r^2 + a^2 - 2ra\cos(180 - \theta)$$

since $\cos(180 - \theta) = -\cos\theta$ we get

$$r_2^2 = r^2 + a^2 + 2ra\cos\theta$$

Neglecting the term $\frac{a^2}{r^2}$ (because $r>>a$)

$$r_2^2 = r^2\left(1 + \frac{2a\cos\theta}{r}\right)$$

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